

Understanding Uncertainty in Data-Driven Research

How Uncertainty Shapes Machine Learning



Understanding Uncertainty in Data-Driven Research

About the course

Unit 1 - Part 1



Who is this course for?



For **researchers**

- who use (or plan to use) **machine learning** and
- want to better understand how **uncertainty shapes**
 - data
 - model outputs
 - research decisions

Course Structure

Unit 1 From Probabilistic Predictions to Decisions

Unit 2 Questioning Ground Truth Labels



Practical Assignment on Ground Truth Labels

Unit 3 Sources of Uncertainty in Machine Learning

Unit 4 Uncertainty When Using Large Language Models



Reflection Checklist

A short word about myself



Cornelia Gruber

- researcher at Department of Statistics at LMU Munich
- statistical approaches for uncertainty in machine learning
 - theory
 - applications in natural language processing and
 - climate science

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Understanding Uncertainty in Data-Driven Research

From Probabilistic Predictions to Decisions

What is uncertainty?

Unit 1 - Part 2



What is uncertainty?

Meaning of **uncertainty** in English



uncertainty

noun [C or U]

UK  /ʌnˈsɜːtɪn.ti/ US  /ʌnˈsɜːtɪn.ti/

Add to word list 

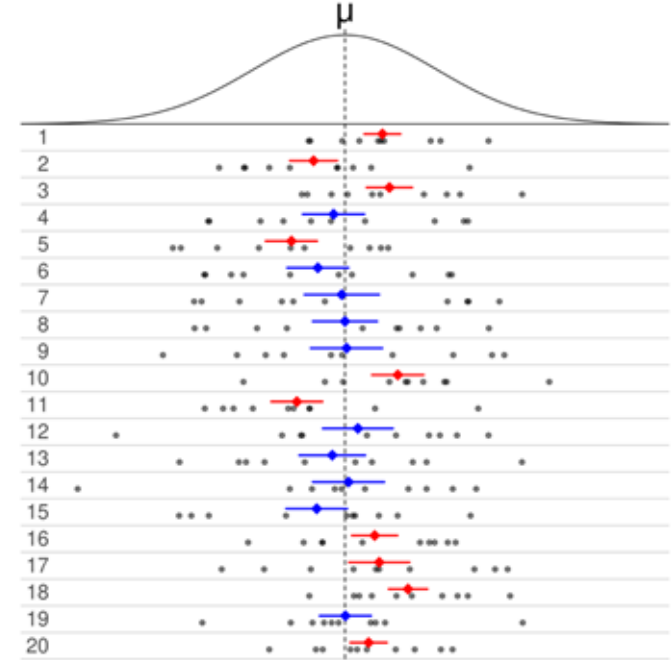
C1

a situation in which something is not known, or something that is not known or certain:

- *Nothing is ever decided, and all the uncertainty is very bad for staff morale.*
- *Life is full of uncertainties.*

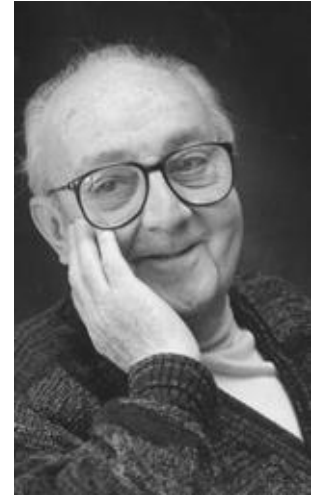
Uncertainty in classical statistics

- p-values, confidence intervals, statistical tests
- rely on distributional assumptions
- include sampling uncertainty



Essentially, all models are wrong, but some are useful.

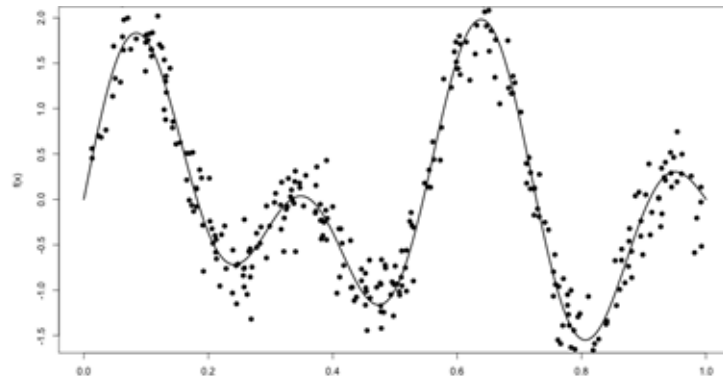
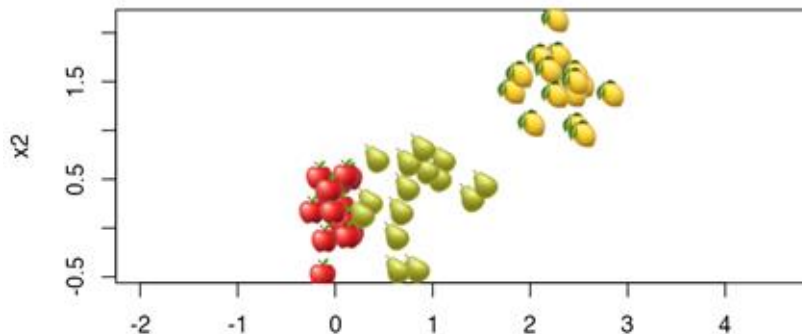
George Box (1919–2013)



Supervised Machine Learning

Goal: learn relationship $f(y|x)$ given input data X and labels Y

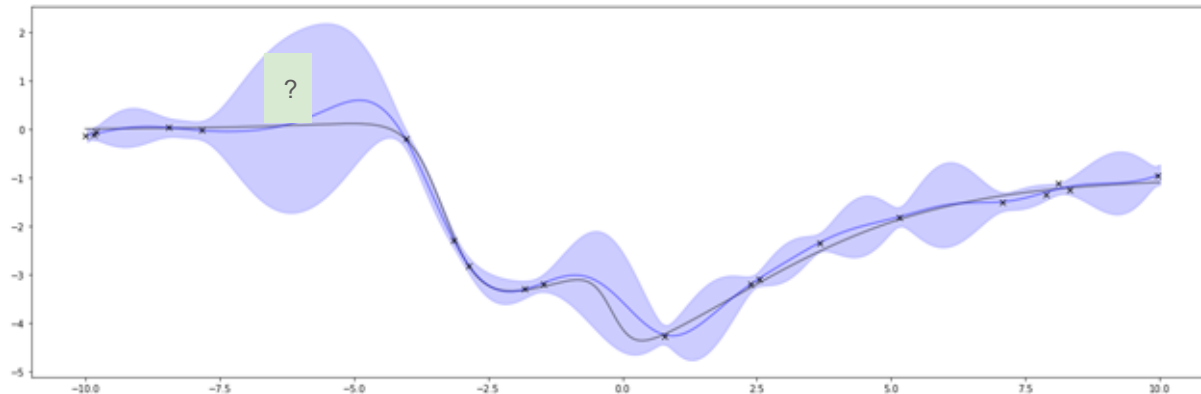
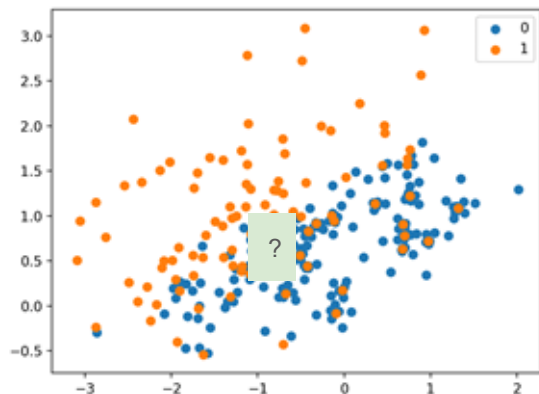
Classification



Machine learning outputs often **look precise**:
a predicted class, score, probability, ranking, or recommendation.

What is uncertainty in machine learning?

- Lack of confidence in a model's prediction
- Likelihood of model's prediction being wrong



Why even care?

A prediction alone does not indicate how much it can be trusted.

Average predictive performance can hide unreliable cases



(a) Covid

(b) Normal

Uncertainty reveals limitations in the data and the model

→ Human in the loop



Uncertainty can itself be scientifically informative



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Understanding Uncertainty in Data-Driven Research

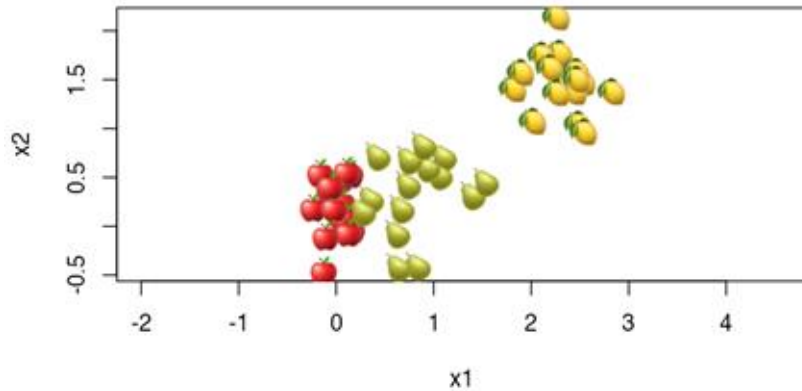
From Probabilistic Predictions to Decisions Calibration

Unit 1 - Part 3

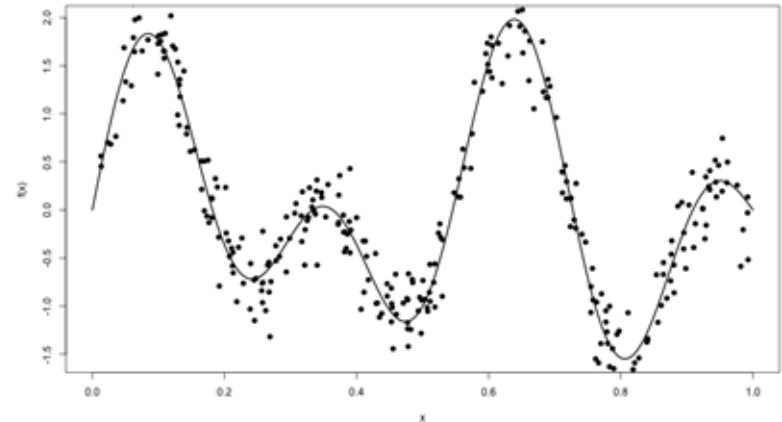


Supervised Machine Learning

Goal: learn relationship $f(y|x)$ given input data X and labels Y

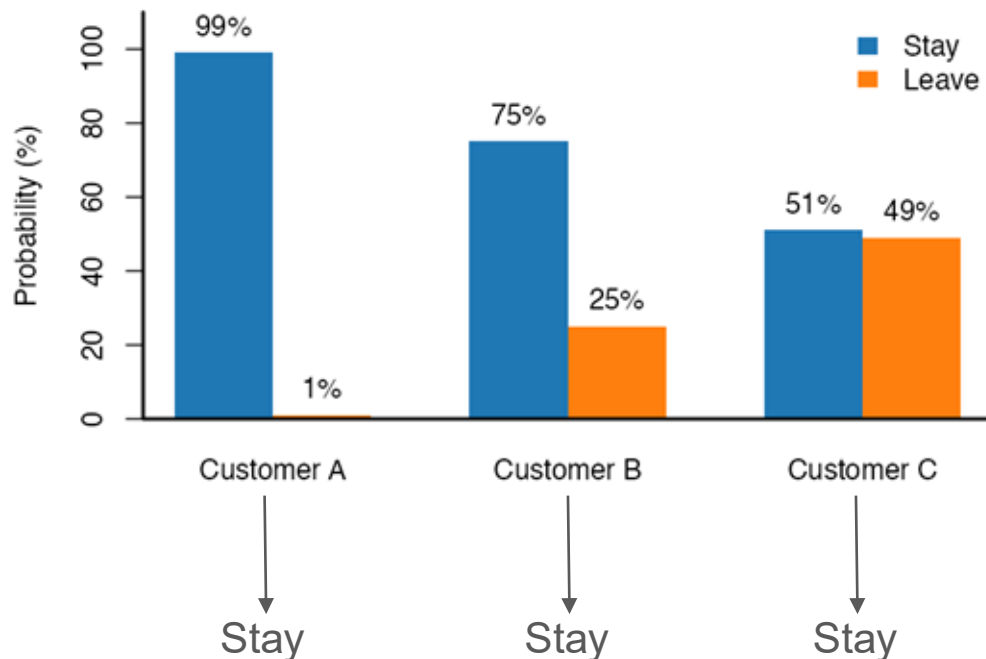


Point prediction: 🍋
prediction: 2.15



Point

Point vs. Probabilistic Predictions



All point predictions are **Stay**.

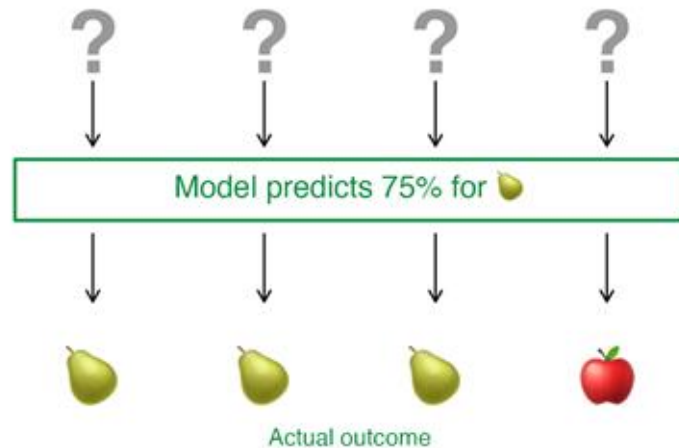
Should all three customers be treated in the same way?

Calibration

Well calibrated:

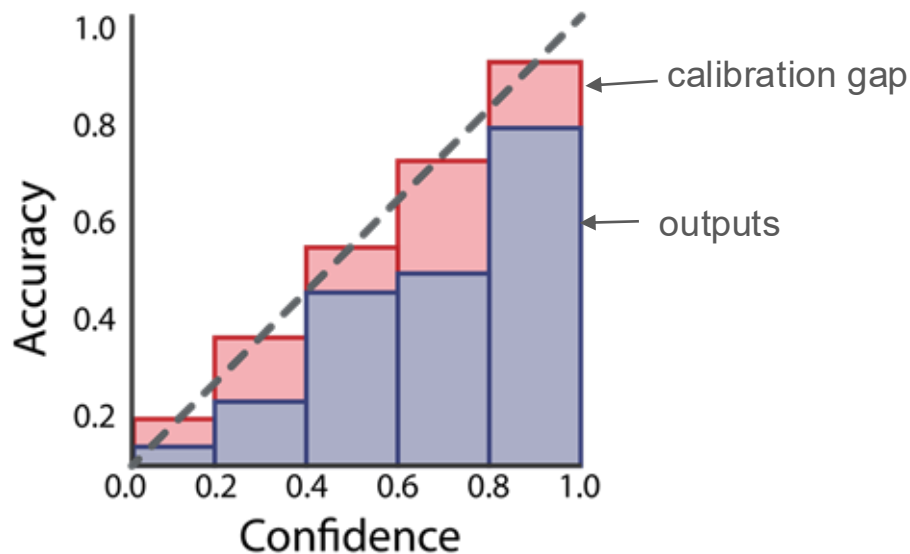
prediction of a class with confidence p is correct $100 * p$ percent of the time

$p(\text{🍌}) = 0.75$, then in 75% of cases $Y = \text{🍌}$



Modern ML algorithms are known to be **miscalibrated**.¹

Calibration and Expected Calibration Error (ECE)



ECE = Weighted Average { , , , ,  }

$$ECE = \sum_{m=1}^M \frac{|B_m|}{N} |Accuracy(B_m) - Confidence(B_m)|$$

CIFAR 10 and CIFAR 100

airplane



automobile



bird



cat



deer



dog



frog



horse



ship



truck



cattle



dinosaur



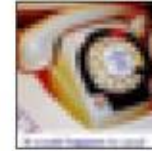
apple



boy



aquarium_fish



telephone



train



cup



cloud



elephant



keyboard



willow_tree



sunflower



castle

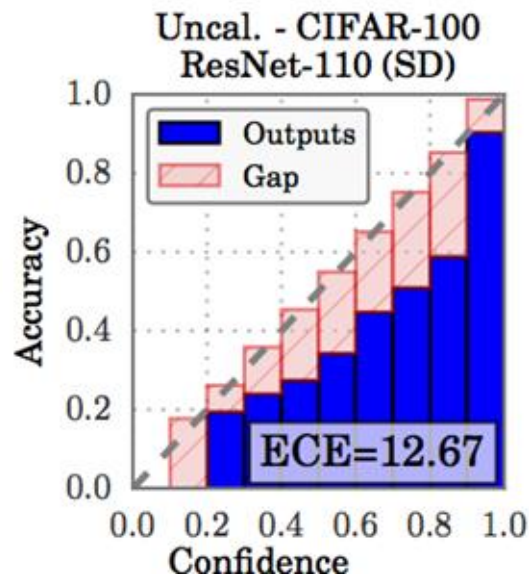


sea



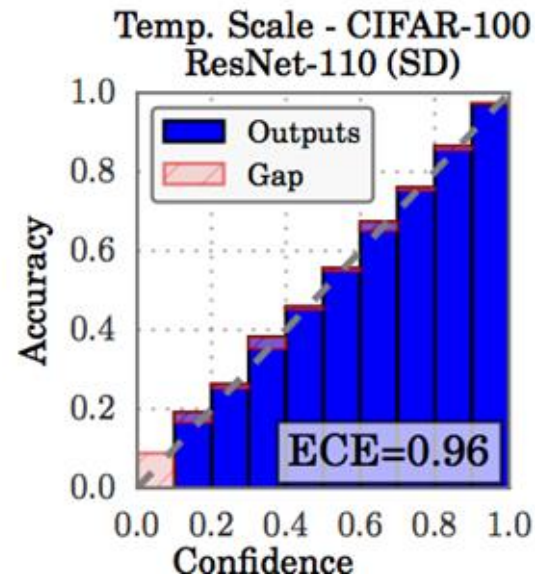
keyboard

Calibration - Temperature Scaling



Temperature Scaling

$$P_i = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)}$$



Why Calibration Matters



- **Probabilities become interpretable:** a prediction with 80% confidence is correct about 80% of the time.
- **Supports risk-aware decisions:** enables thresholding based on meaningful probabilities rather than arbitrary confidence scores.
- **Improves comparability across models:** calibrated probabilities can be directly compared, combined, or averaged.
- **Enables uncertainty communication:** calibrated probabilities provide an intuitive way to communicate confidence to users and stakeholders.
- **Foundation for many uncertainty-aware methods:** calibration is often a prerequisite for decision support, active learning, and probabilistic forecasting.

Caveats of Calibration

- **Calibration is a population property:** it guarantees reliable probabilities on average, not for individual predictions.
- **Well-calibrated $\neq$ good model:** a model can be perfectly calibrated while having poor accuracy or little predictive value (e.g., always predicting the class frequency).
- **Calibration does not identify difficult cases:** it does not ensure that uncertain predictions correspond to mistakes. Detecting errors requires good discrimination, not just calibration.
- **Confidence calibration is only one notion of calibration:** multiclass settings, class-wise calibration, and label distributions (e.g., human disagreement) often require different calibration concepts.
- **Evaluation is challenging:** Expected Calibration Error (ECE) depends on arbitrary binning, ignores most class probabilities, and different models can have identical ECE despite different behavior.

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Understanding Uncertainty in Data-Driven Research

From Probabilistic Predictions to Decisions

Entropy

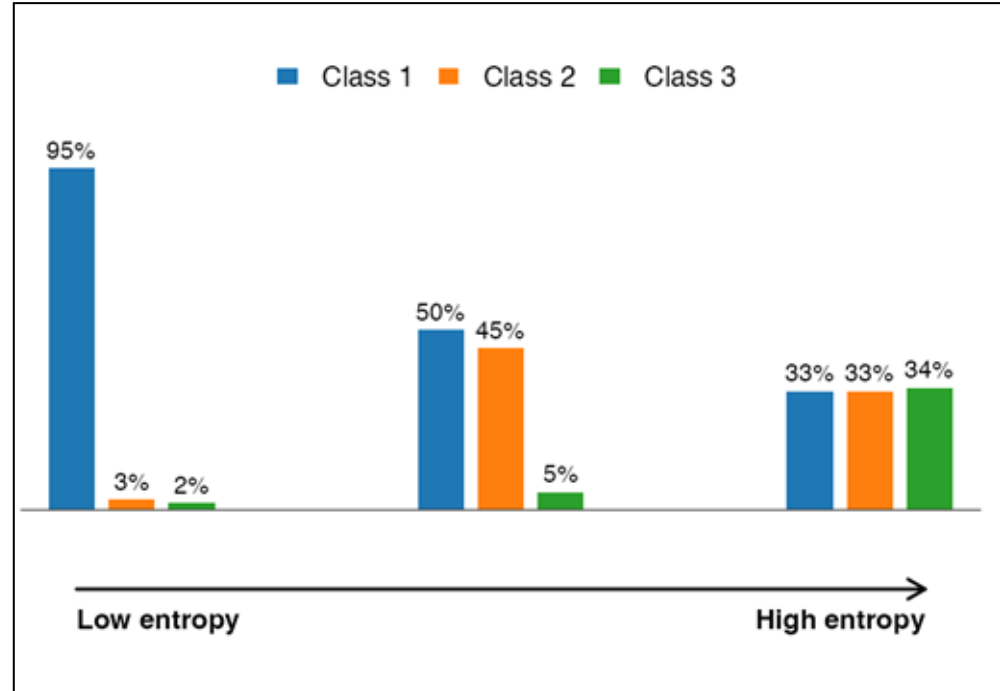
Unit 1 - Part 4



Entropy

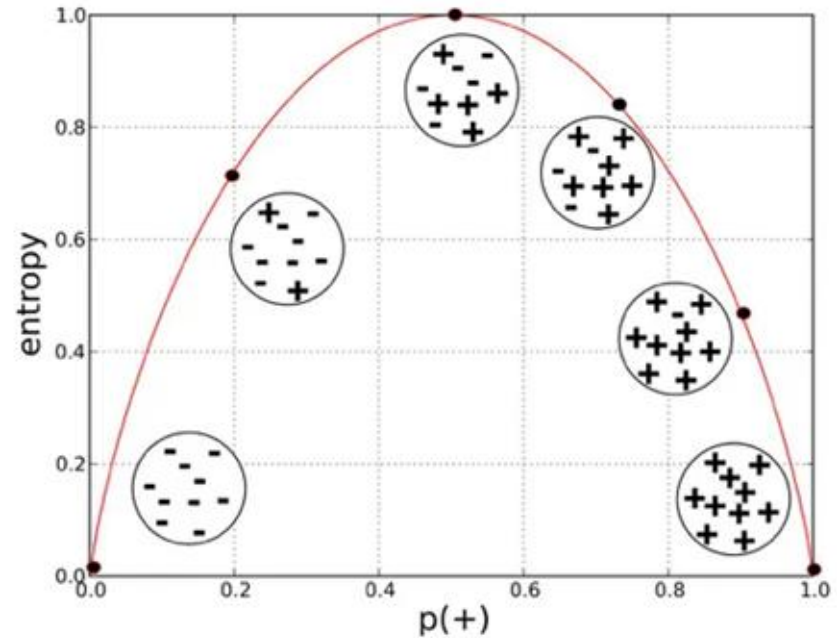
Measures the amount of uncertainty, randomness, or "surprise" in a dataset.

- **low**: probability is concentrated
- **high**: probability is spread across several classes
- **maximal** for a uniform distribution



Entropy

$$H(X) = - \sum_i p_i \log_2(p_i)$$



Entropy



Can tell us

- how dispersed a predicted class distribution is
- whether one class dominates
- which cases are relatively ambiguous
- how uncertainty changes across observations

Cannot tell us by itself

- whether the prediction is correct
- whether probabilities are calibrated
- why the model is uncertain
- whether an observation is out of distribution
- which decision should be taken

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Understanding Uncertainty in Data-Driven Research

From Probabilistic Predictions to Decisions Conformal Prediction

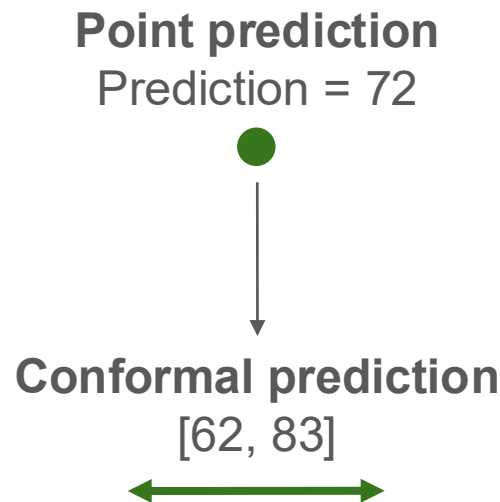
Unit 1 - Part 5



Conformal Prediction

- Reliable uncertainty sets for **any black-box** machine learning models
- Key idea: coverage you can audit:
covers the true value about 90% of the time
- Predicts sets/intervals, not certainty scores

**Intuition: Use a held-out calibration set to ask,
"How large must our intervals/sets be so that past
errors would have been covered?"**



Conformal Prediction



1. Split Data

Keep calibration data separate from training

2. Train Model

Fit any model: NN, RF, XGBoost, regression, classifier

3. Calibrate Errors

Compute non-conformity scores on held-out cases

4. Predict Set

Choose interval/label set using the empirical quantile

Marginal coverage guarantee: $P(Y \in C(X)) \geq 1 - \alpha$

probability that true label Y is in conformal interval/set is at least $1 - \alpha$ (e.g., 90%)

Only assumption: training and test data is **exchangeable**

→ they are drawn from the same distribution and their order does not matter

Conformal Regression and Classification

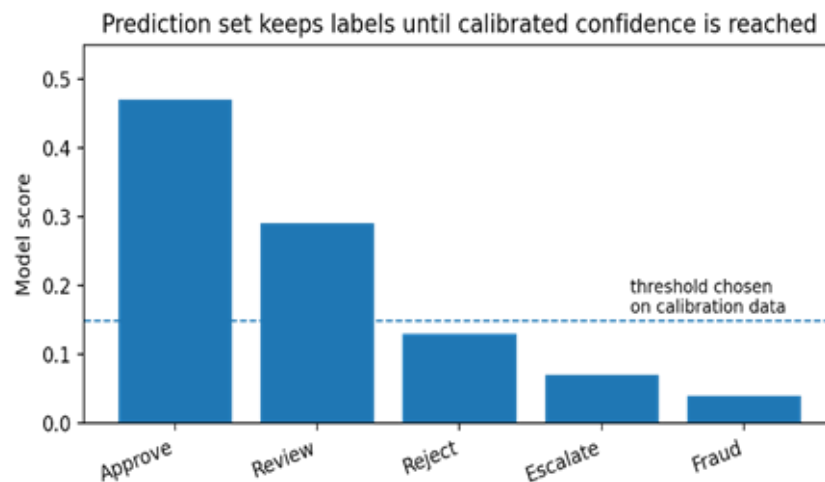
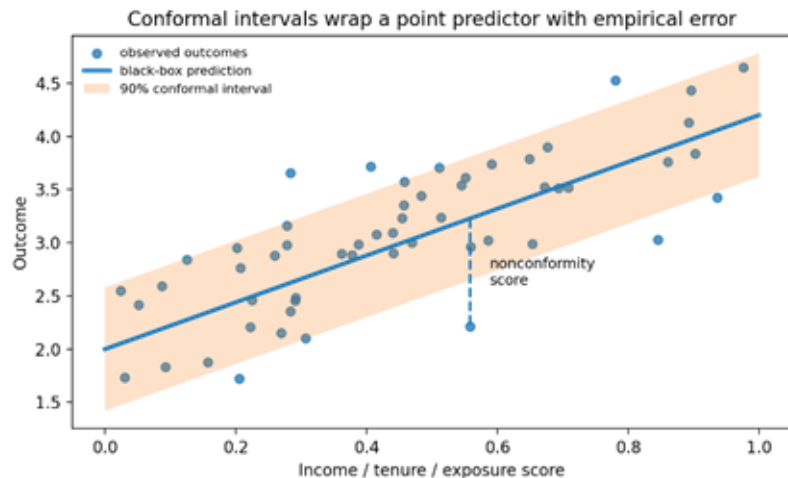


Score = absolute residual $|y - \hat{y}|$

Interval = $\hat{y} \pm \hat{q}$

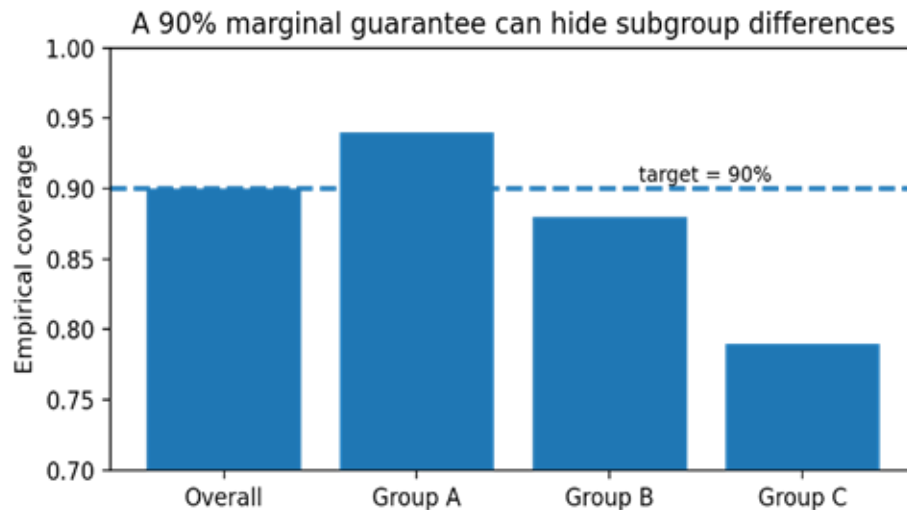
Score = One minus probability $1 - p(y|\hat{x})$

Set = all classes below quantile threshold



Caveats

- Coverage is usually marginal: subgroups may be under-covered
- Guarantees rely on comparable calibration and deployment data
- Intervals can be valid but too wide to support a decision.
- It quantifies prediction uncertainty, not causal identification.



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Understanding Uncertainty in Data-Driven Research

From Probabilistic Predictions to Decisions

Error costs and meaningful human oversight

Unit 1 - Part 6

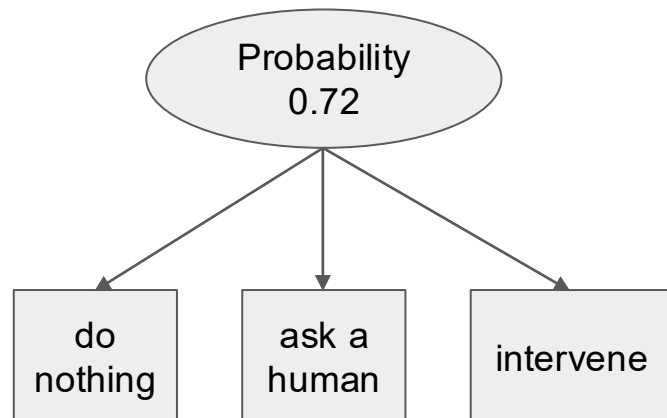


A prediction is not a decision

A model estimates what may happen.

A decision rule determines what happens next.

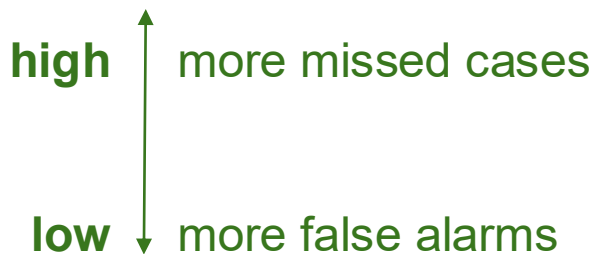
Today's question: Who sets that rule and who bears the cost when it is wrong?



The same score can lead to different actions

The score does not contain the **costs, rights, rules, or values** needed for a decision.

Every **threshold** exchanges one kind of error for another.



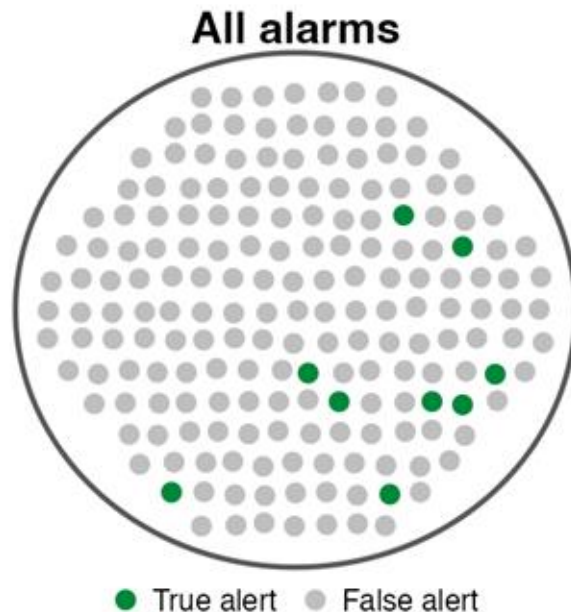
Predicted probability	Context	Plausible Action
10%	Spam email	Move to spam
10%	Serious disease	Order a follow-up test
10%	Welfare fraud	Do not sanction, review

Rare events and false alarms

Imagine 180,000 train station visitors per day:

- 10 people sought by system
- 90% sensitivity $\rightarrow$ 9 true alerts
- 0.1% false-positive rate $\rightarrow$ 180 false alerts

Only about **1 in 21 alerts** would identify a sought person.



A threshold is a policy choice

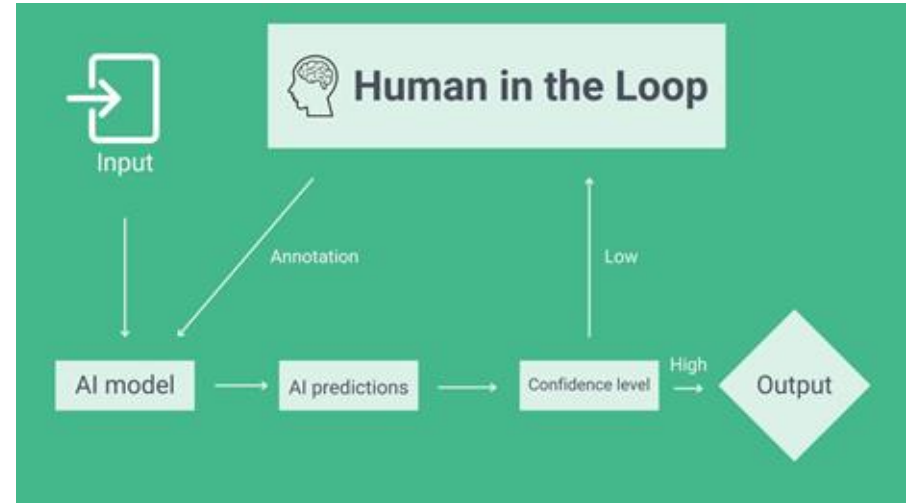
1. What **action** does crossing the threshold trigger?
2. How **harmful** and **reversible** is each error?
3. Who receives the **benefit** and who bears the **burden**?
4. What **legal** or evidentiary standard applies?
5. Is there a **safer intermediate action**?



Human-in-the-Loop

A human actively participates in the **operation, supervision or decision-making** of an automated system

- accuracy
- safety
- accountability
- ethical decision-making.



Caveats of Human-in-the-Loop

- **Over- and under-reliance:** People may follow incorrect model advice or ignore useful recommendations
- **Decision variability:** Expertise, incentives, workload, and risk preferences affect how predictions are translated into action
- **Prediction shapes judgment:** Model outputs can anchor attention and influence which factors humans consider



Caveats of Human-in-the-Loop

- **Selective outcomes:** Human interventions determine which outcomes are observed, creating biased data for evaluation and retraining
- **Feedback loops:** Decisions change the environment and future data, reinforcing earlier model-human choices
- **Unclear accountability:** Responsibility becomes difficult to assign when outcomes result from joint human-model decisions

**A human in the loop
does not automatically
make an ML system
safer, fairer, or more
accurate.**

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